

CMPT 354 Fall 2023

Database Systems I

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Assignment 3

Maximum marks: 100 (5% of the total marks)

Due date: October 15, 2023

The goal of this assignment is to create a SQL Server database for managing the Yelp data for Edmonton. We provide some sample data based on the dataset available on

<https://www.yelp.com/dataset>.

Using your assigned database on the Cypress server, you will create the tables and constraints and insert some sample data available on CourSys.

Please, check the following link for instructions on how to use Microsoft SQL Server in CSIL:

<http://www.sfu.ca/computing/about/support/csil/windows/how-to-use-sql.html>

This link also presents instructions on how to import data from a file into your database:

<https://www.sfu.ca/computing/about/support/csil/windows/how-to-use-sql.html#import-export>

Assignment 3.1

Use SQL to create the tables and constraints in your SQL Server database according to the description below.

Your database should have the following tables, where the primary key is underlined, and foreign keys are shown in *italic*. You must use the exact names and domains specified. Marks will be deducted if you fail to do so.

Tables

business = (business_id, name, address, city, postal_code, stars, review_count)

checkin = (checkin_id, *business_id*, date)

tip = (tip_id, *user_id*, *business_id*, date, compliment_count)

review = (review_id, *user_id*, *business_id*, stars, useful, funny, cool, date)

user_yelp = (user_id, name, review_count, yelping_since, useful, funny, cool, fans, average_stars)

friendship = (user_id, friend)

Attributes

You should use the domains listed below. Consider the described constraints and primary keys.

Table business

- business_id: string(22), primary key.
- name: variable-length string(60), not null.
- address: variable-length string(75).
- city: variable-length string (30), not null.
- postal_code: variable-length string(7).
- stars: decimal(2, 1), in [1,5] interval.
- review_count: integer, default: 0, non-negative.

Table checkin

- checkin_id: integer, primary key.
- business_id: string(22), not null.
- date: date and time, not null, default: current date.

Table tip

- tip_id: integer, primary key.
- user_id: string(22), not null.
- business_id: string(22), not null.
- date: date and time, not null, default: current date.
- compliment_count: integer, default: 0, non-negative.

Table friendship

- user_id: string(22), primary key
- friend: string(22), primary key

Table review

- review_id: string(22), primary key.
- user_id: string(22), not null.
- business_id: string(22), not null.
- stars: integer, not null, between 1 and 5.
- useful: integer, default: 0, non-negative.
- funny: integer, default: 0, non-negative.
- cool: integer, default: 0, non-negative.
- date: date and time, default: current date.

Table user_yelp

- user_id: string(22), primary key.
- name: variable-length string(35), not null.
- review_count: integer, default 0, non-negative.
- yelping_since: date and time, not null, default: current date.
- useful: integer, default: 0, non-negative.
- funny: integer, default: 0, non-negative.
- cool: integer, default: 0, non-negative.
- fans: integer, default: 0, non-negative.
- average_stars: decimal(3, 2), in [1,5] interval.

Note: Consider a fixed-length string where not specified otherwise.

Constraints

Your database should enforce the primary key constraints noted above.

The following foreign key constraints also need to be enforced:

- checkin.business_id references business.business_id
- tip.business_id references business.business_id
- tip.user_id references user_yelp.user_id
- review.business_id references business.business_id
- review.user_id references user_yelp.user_id
- friendship.user_id references user_yelp.user_id
- friendship.friend references user_yelp.user_id

The response to the violation of foreign keys constraints should be to not allow deletion or update of referenced records.

Assignment 3.2

Insert the sample data provided on CourSys into the tables that you have created. The zip file “assignment3.zip”, posted on CourSys under Datasets, contains one file for each table.

Use one of the following two options:

1) Command prompt: bcp

```
bcp "dbName.dbo.tableName" in "ifile" -S SQLserver -b 1 -m 100 -T -c -F 1
```

2) T-SQL: bulk insert

Example:

```
"BULK INSERT dbo.business FROM 'd:\userdata\...\business.csv' WITH (fieldterminator=',',  
rowterminator='\n', firstrow=2)"
```

Remember you will need to insert data in a sensible order that respects the foreign key constraints. Report the order in which you execute the statements.

Submission

Create the tables and constraints in your assigned database on the Cypress server and import the data provided. Your database must contain an exact copy of the provided data without modifications (insertions, deletions, updates).

Submit a pdf-file Assignment3.pdf with

- all your SQL statements to create tables,
- all your SQL statements to create constraints,
- and the order of tables when inserting the data to CourSys.